Embedded Implementation and Analysis of Post-Quantum Signature Schemes (CROSS, LESS) on STM32F4

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1 Introduction

Cryptography plays a fundamental role in modern digital systems and is essential for ensuring the confidentiality and integrity of data in everyday applications. From online communication to financial transactions, secure cryptographic schemes are required to protect sensitive information. With the rapid development of quantum computing, many classical cryptographic schemes are expected to become insecure in the future. This creates the need for post-quantum cryptography (PQC), which aims to provide secure alternatives that remain resistant against quantum attacks.

To address this challenge, the National Institute of Standards and Technology initiated a standardization process to evaluate and select suitable PQC algorithms [1]. In addition to the main standardization effort, further digital signature schemes have been proposed in subsequent calls to explore alternative approaches and increase diversity.

One of these candidates is CROSS, a code-based signature scheme based on the hardness of the Restricted Syndrome Decoding Problem (R-SDP) [2]. The scheme is constructed by transforming an interactive zero-knowledge identification protocol into a non-interactive signature scheme using the Fiat–Shamir transform [2]. Code-based cryptography is considered a promising approach due to its long-standing resistance against known classical and quantum attacks.

In addition to CROSS, other code-based signature schemes such as LESS [3] have been proposed within the NIST additional signature call, providing alternative design approaches and performance characteristics.

CROSS aims to provide efficient digital signatures with competitive performance while maintaining strong security guarantees [2]. Similar design goals can also

be observed in other code-based candidates such as LESS.

While reference implementations of these schemes are available in C, deploying such algorithms on constrained embedded systems remains a challenging task. Limited memory, computational resources, and potential side-channel leakage must be taken into account. Therefore, this project focuses on implementing selected components of multiple PQC signature schemes on an embedded platform and preparing them for side-channel evaluation.

2 Goals

The main goal of this project is to implement and analyze selected components of post-quantum signature schemes on an embedded platform (Riscure Pinata with STM32F4), starting with CROSS, LESS and extending to additional schemes if time permits.

Rather than implementing full algorithms, the focus lies on adapting critical subroutines to a constrained environment and preparing them for side-channel evaluation.

To achieve this, the project is structured into the following objectives:

2.1 Understanding the Algorithm

- Study the structure and underlying principles of CROSS and LESS
- Identify relevant subroutines suitable for embedded implementation

2.2 Embedded C Implementation

- Adapt and implement selected parts of CROSS and LESS in C
- Port the implementation to the STM32F4 microcontroller
- Ensure compatibility with constrained memory and execution environment

2.3 Validation and Testing

- Verify correctness by comparing outputs with the reference implementation
- Develop test cases for individual subroutines

2.4 Python Integration and Tooling

- Compile the C implementations into shared libraries
- Interface the implementations with Python for testing and automation
- Enable calculation of intermediate values for validation purposes
- Integrate with ChipWhisperer for data input/output handling

2.5 Preparation for Side-Channel Analysis (SCA)

If time permits, the project will also consider the following aspects:

- Prepare the implementation such that side-channel measurements become feasible
- Identify suitable subroutines for future analysis
- Estimate execution times of selected operations
- Evaluate the feasibility of SCA based on runtime and sampling constraints

2.6 Performance and Feasibility Evaluation

- Measure the runtime of selected subroutines on the STM32F4 microcontroller
- Analyze memory consumption and identify potential bottlenecks
- Evaluate whether the implemented components meet the constraints of the embedded platform
- Estimate the feasibility of executing larger parts or the full algorithm on the target device
- Assess the suitability of selected operations for side-channel analysis based on execution time and expected trace length

3 Timeline of our work

Milestone	Timeline
Background study and analysis	Week 1-3
Selection of subroutines	Week 4
Python & C implementation (CROSS, LESS)	Week 5-15
Validation (CROSS, LESS)	Week 11-12
Python & C implementation (other schemes)	Week 13-20
Validation (other schemes)	Week 21-22
Early timing analysis (if time allows)	Week 18-22
Writing, documentation, preparing the presentation and buffer for delays	Week 23-26
Submit work, presentation, report	Week 27

References

[1] National Institute of Standards and Technology. *Post-Quantum Cryptography Standardization*. 2025. URL: `https://csrc.nist.gov/projects/post-quantum-cryptography`.

[2] CROSS Team. *CROSS Specification v2.2*. 2025. URL: `https://www.cross-crypto.com/CROSS_Specification_v2.2.pdf`.

[3] LESS Team. *LESS: Linear Equivalence Signature Scheme*. 2025. URL: `https://www.less-project.com/LESS-2025-02-07.pdf`.